Prior Sensitivity in Bayesian Logistic Regression: A Small-Sample Medical Study

William Chen
University of California, Irvine
Instructor: Prof. Weining Shen
Stat 205P: Bayesian Data Analysis

Abstract

In Bayesian logistic regression, the choice of prior is inconsequential when data are plentiful but decisive when they are scarce. We demonstrate this using the `birthwt` dataset ($n = 189$), comparing three priors (diffuse $\mathcal{N}(0, 100)$, weakly informative $\mathcal{N}(0, 2.5)$, and a clinically informative prior) across the full sample and subsamples of $n \in \{20, 40, 80\}$. At full data all three are interchangeable (AUC ≈ 0.747). Under small samples they are not: at $n = 20$ the diffuse prior collapses under complete separation ($\hat{\beta}_{\texttt{smoke1}} = 203$) while the regularizing priors stay stable. Crucially, how quickly a coefficient escapes prior sensitivity is governed not by sample size but by the number of positive cases for that predictor: smoking stabilizes by $n = 40$, whereas hypertension (only 6.3% of patients) remains unstable until $n \approx 80$. For small clinical datasets the recommendation is concrete: prefer a weakly informative prior to a diffuse one, and treat rare predictors with particular caution, since they are the last to shed the prior's influence.

1 Introduction

In Bayesian analysis the prior's influence is not constant: when data are abundant the likelihood dominates and the prior fades into the background, but when samples are scarce it can substantially shape, and even distort, the posterior. This matters most in clinical research, where small samples are the rule rather than the exception.

This report presents a prior sensitivity analysis of Bayesian logistic regression using the `birthwt` dataset ($n = 189$), comparing three priors that span from diffuse $\mathcal{N}(0, 100)$ through weakly informative $\mathcal{N}(0, 2.5)$ to a clinically informative prior. Refitting each across sample sizes ($n \in \{20, 40, 80, 189\}$), we find that the priors are interchangeable at full data but diverge sharply at small n, and that this sensitivity is governed not by sample size alone but by predictor rarity, with direct implications for clinical datasets where the most important risk factors are often uncommon.

2 Data

2.1 Dataset Overview

The `birthwt` dataset ($n = 189$) was collected at Baystate Medical Center, Springfield, MA (1986). The binary outcome is low birth weight (< 2.5 kg), with 130 normal and 59 low birth weight cases (31.2% positive rate). All analyses use `stan_glm` (`rstanarm`), fit with 4 chains $\times$ 1000 iterations.

2.2 Predictors

Predictors include maternal age (`age`), pre-pregnancy weight (`lwt`), race (3-level factor), smoking status during pregnancy (`smoke`), prior preterm labor (`ptl`), hypertension history (`ht`), uterine irritability (`ui`), and first-trimester physician visits (`ftv`). Continuous features show low pairwise correlations (max $|r| = 0.22$), suggesting no multicollinearity concerns. Among the binary predictors, the positive class is uncommon and unevenly distributed: smoking is present in 74 of 189 mothers, uterine irritability in 28, and hypertension in only 12. As the subsampling results show, this spread becomes consequential under small samples.

3 Methods

3.1 Prior Specification

Three prior configurations were evaluated:

Model	Coefficients	Intercept
Diffuse	$\mathcal{N}(0, 100)$	$\mathcal{N}(0, 100)$
Weak	$\mathcal{N}(0, 2.5)$	$\mathcal{N}(0, 10)$
Informative	`smoke`, `ht`: $\mathcal{N}(1, 0.5)$; others $\mathcal{N}(0, 1)$	$\mathcal{N}(0, 5)$

The informative prior encodes the clinical knowledge that smoking and hypertension are established risk factors for low birth weight. All other predictors receive a weakly informative $\mathcal{N}(0, 1)$ prior.

3.2 Analyses

Three analyses were conducted:

1. **Prior sensitivity at full data** ($n = 189$): posterior coefficient comparison, LOO model comparison, and AUC.

2. **Subsampling experiment**: all three models were refit at $n \in \{20, 40, 80, 189\}$ to track posterior mean convergence and credible interval width across sample sizes.

3. **Predictive performance**: AUC was computed at each sample size and prior to assess whether informative priors improve prediction under small n.

Convergence was verified for all models via $\hat{R} \leq 1.002$ and visual inspection of trace plots.

4 Results

4.1 Full Data ($n = 189$): Priors Have Little Effect

At $n = 189$, all three models produce nearly identical posterior estimates and converge well. Table 1 reports posterior means for the two clinically relevant predictors, and Figure 1 shows the full coefficient comparison across all predictors.

Variable	Diffuse	Weak	Informative
`smoke1`	0.983	0.933	0.915
`ht1`	1.994	1.827	1.311

Table 1: Posterior means (log-odds) for the two clinically targeted predictors at $n = 189$. These are the variables for which the informative prior encodes directional clinical knowledge.

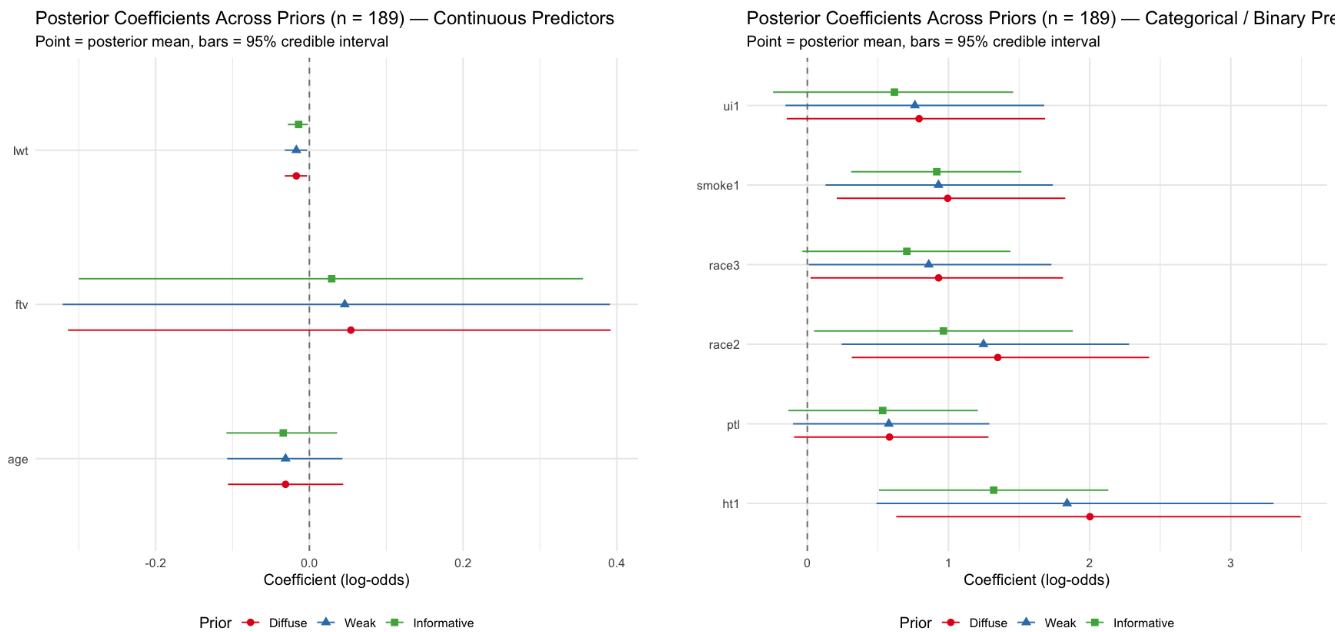

Figure 1: Posterior coefficients across priors at $n = 189$. Left: continuous predictors. Right: categorical/binary predictors. Points = posterior mean; bars = 95% credible interval.

The informative prior modestly shrinks `ht1` (from 1.994 to 1.311), but this difference is small in practical terms. Table 2 reports LOO and AUC for all three models; the differences are negligible, confirming that prior choice has little practical effect at $n = 189$.

Model	LOOIC	AUC
Diffuse	223.85	0.746
Weak	223.32	0.747
Informative	220.43	0.748

Table 2: LOO model comparison and AUC at full data ($n = 189$). Lower LOOIC indicates better fit.

With 189 observations, the likelihood dominates and prior choice is inconsequential.

4.2 Subsampling: Prior Sensitivity at Small n

To examine how prior influence changes with sample size, we refit all three models at $n \in \{20, 40, 80, 189\}$. We focus on `smoke1` and `ht1`, the two predictors for which the informative prior encodes directional knowledge, as these show the largest divergence across priors at small n (Figure 2).

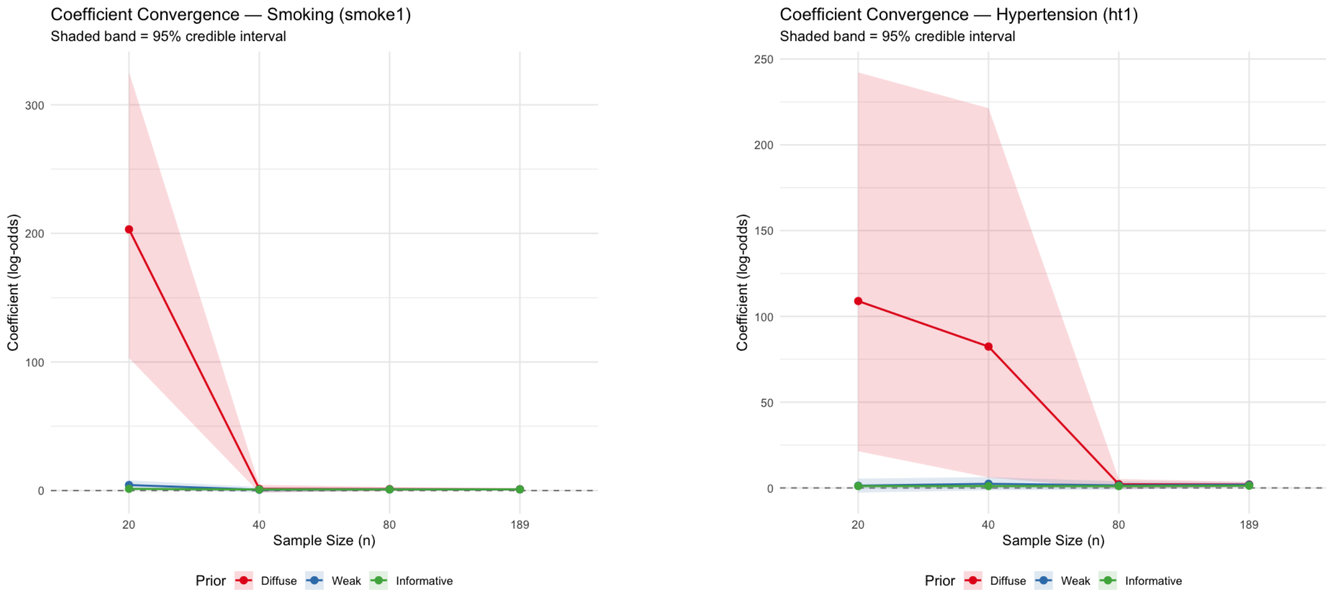

Figure 2: Posterior mean convergence across sample sizes. Left: `smoke1`. Right: `ht1`. Shaded bands = 95% credible intervals.

Under the diffuse prior, complete separation produces implausible estimates at $n = 20$ ($\hat{\beta}_{\texttt{smoke1}} = 203$, $\hat{\beta}_{\texttt{ht1}} = 109$). `smoke1` recovers by $n = 40$, but `ht1` remains unstable ($\hat{\beta} = 82.4$): hypertension affects only 12 of 189 patients (6.3%), so small subsamples rarely contain enough `ht` = 1 cases to resolve separation. By $n = 80$, all models stabilize. The weak and informative priors remain in clinically plausible ranges throughout.

4.3 Predictive Performance

We assess whether prior choice affects predictive performance, and whether a more informative prior offers any advantage at small n (Figure 3).

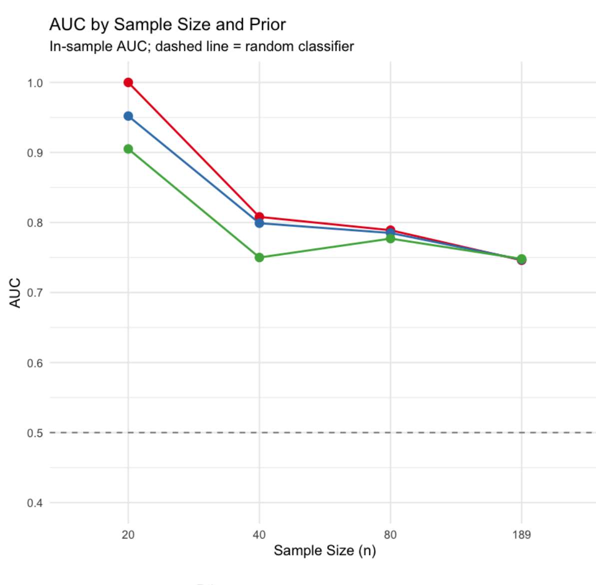

Figure 3: In-sample AUC by sample size and prior. Dashed line = random classifier baseline (AUC = 0.5).

The diffuse prior achieves AUC $= 1.0$ at $n = 20$, a direct consequence of complete separation: a model that perfectly separates the training data is not a good model, it is an overfit one. The informative prior, while showing lower in-sample AUC at small n (0.905 at $n = 20$), produces more stable and trustworthy estimates. By $n = 189$, all three priors converge to AUC ≈ 0.747.

5 Discussion

The prior is not neutral at small n. "Letting the data speak" assumes the data are loud enough to speak; at $n = 20$ they are not, and a diffuse prior actively destabilizes inference instead of staying out of the way.

Rarity acts as a threshold, not a gradient. What decides when a coefficient becomes trustworthy is not the sample size but the number of positive cases for that predictor. At $n = 20$ every coefficient is unstable: even smoking, the most common predictor, cannot escape separation in so small a subsample. By $n = 40$ only hypertension remains affected. Uterine irritability, though rarer than smoking (28 versus 74 positive cases), recovers just as quickly, so the effect is not a smooth function of rarity but a threshold: with only 12 positive cases in the full data, hypertension's subsamples rarely contain enough `ht` $= 1$ observations to resolve separation.

Limitations. AUC is reported in-sample and likely overstates predictive performance, especially at small n. The subsampling experiment uses a single fixed random subset per sample size; averaging over multiple draws would give a more stable picture of convergence thresholds.

6 Conclusion

Prior choice matters most when data are scarcest. At $n = 20$ a diffuse prior produces unstable, uninterpretable posteriors while a weakly informative prior restores stability; by $n = 189$ the differences are negligible. This threshold is not uniform, however: rare predictors such as hypertension stay prior-sensitive longest, since stability depends on the number of positive cases, not the sample size alone.

For small clinical datasets the takeaway is direct: make a weakly informative prior the default rather than a diffuse one, and treat rare predictors with particular caution, since they are the last to shed the prior's influence.